

Chapter Four

Defining problem and research objectives

4.1. Defining problem

- ❖ Like *a medical doctor*, a researcher must examine all the symptoms or presented to him or observed by him concerning a problem before *he can diagnose* correctly.
- ❖ There is a famous saying that “*problem well defined is half solved*”. This is because of the problem is stated properly, the objectives *will not be clear*, once objectives are not clearly defined, *data collection becomes meaningless*. To define the problem correctly the researcher must know what a problem is.

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- ❖ *Research problem* is refers to some difficulty which a researcher t experiences in context of to either theoretical or practical situation or wants to obtain solution.
- ❖ *It is an issue in need of investigation.*
- ❖ *A good problem statement* should be clearly written and be able to immediately capture the readers.
- ❖ It should be able to advance knowledge or be of interest to society.
- ❖ In addition, it should indicate *the scope of the study* and be objectively researchable.

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- ❖ The major problems most managers facing is how to convert a management problem in to a researchable problem.
- ❖ It is *not hard* to define management problem, but it is difficult to identify single dilemma on which to focus.
- ❖ Choosing a wrong or *incorrectly defined management problem will* result in waste of resource as well as lead to *wrong decisions* costing further on the company.

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Example: one of the largest cinema chains in uk faced with a problem of declining audiences. The team in the first meeting came up with initial problem statement "to discover why cinema audiences are declining? But some of the members are not agreed on this so again they propose an alternative statement as *'to identify the ways in which more people could be attracted to attend the cinema.*

❖ *From the above paragraph which one is managers problem and research problem?*

Types of research

- ❖ Research can be *classified in different ways: basic and applied; qualitative and quantitative; conceptual and empirical; etc...* _
- ❖ *Applied Research* (also called action research): The purpose of applied research is to solve an immediate, practical problem. It is oriented to a specific problem.
- ❖ *Basic Research* (also called *pure or fundamental research*): It has a more general orientation, adding to the existing body of knowledge in the discipline

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- ❖ *Quantitative Research*: Research based on the measurement of quantity or amount.
- ❖ *Qualitative Research*: Research concerned with qualitative phenomenon.
- ❖ *Conceptual Research*: related to *abstract ideas* or *theories*.
Commonly used by *philosophers and thinkers*.
- ❖ *Empirical Research*: relies on experience or observation alone. It is data based research *coming up with conclusions* which are capable of being verified by *observation or experiment*.

Chapter Five

Research design

Introduction

- ❖ *Research design* provides the glue that holds the research project together.
- ❖ *A design* is used to structure the research, to show *how all of the major parts* of the research project like:
- ❖ The *samples or groups, measures, treatments or programs, and methods of assignment* -- work together to try to address the central research questions.

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- ❖ *According to Green and Tull: A Research Design* is the specification of methods and procedures for acquiring the information needed.
- ❖ It is *the over-all operational pattern or framework* of the project that stipulates *what* information is to be collected from *which* sources by what procedures.
- ❖ Hence it is clear that *Research design* is the *blueprint* for research. It lays down the methodology involved in the collection of information and arriving at meaningful conclusions from the same.

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- ❖ There are *many classifications* accepted for a Research Design.
- ❖ One of the *most accepted classification* is grouping it under *three* types:
 - *Exploratory research design*
 - *Descriptive research design*
 - *Causal research design*
- ❖ As the term suggests, *exploratory research* is often conducted because a *problem has not been clearly* defined as yet, or its real scope is as yet unclear.

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- ❖ It is a process of discovery wherein you *uncover* as many ideas as possible.
- ❖ It allows the researcher to *familiarize him/herself* with the problem or concept to be studied, and perhaps generate hypothesis to be tested.
- ❖ It expands knowledge. It is the initial research, before more conclusive research is undertaken.
- ❖ *Exploratory research* relies more on secondary data.

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- ❖ The *results of exploratory research* are not usually useful for decision-making by themselves, but they can provide significant insight into a given situation. The findings of this can be used to develop the *research further*.
- ❖ *Descriptive research design*: describes the "who, what, when, where and how" of a situation and *not what caused* it.
- ❖ Therefore, *descriptive research* is used when the objective is to provide *a systematic description* that is as factual and accurate as possible.

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- ❖ *Causal research* is undertaken to see if there *is a cause and effect* relationship between variables.
- ❖ In order to determine causality, it is important *to hold* the variable that is assumed *to cause the change* in the other variable(s) constant and then measure the *changes in the other variable(s)*.

Sampling design

- ❖ All items in any field of inquiry constitute a ‘universe’ or a ‘population’. *A complete enumeration* of all items in the population is known as a census inquiry. In such an inquiry, when all items are covered, *no element of chance* is left and highest accuracy is obtained.
- ❖ This type of inquiry, however, involves *a great deal of time, money and energy*. Therefore, *when the field of inquiry is large*, this method becomes difficult to adopt because of the resources involved. At times, this method is beyond the reach of ordinary researchers.

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- ❖ *Government, in very rare cases*, is the only institution which can get the complete enumeration carried out. *For example, population census in our country is carried out once in a decade.*
- ❖ Sometimes it is possible to *obtain sufficiently accurate* results by studying only a part of total population, technically called *samples*.
- ❖ The process of selecting samples is called *sampling technique*.
- ❖ The samples selected should be *as representative* of the total population.

THE NEED FOR SAMPLING

The following points summarize the benefits of studying samples:

- ✓ There could be *resource (time, finance, manpower, etc.) limitations* which would make it difficult to study the whole population.
- ✓ In some cases, tests may be *destructive*.
- ✓ Sampling is the only process possible if the population is infinite.
- ✓ Sampling provides much quicker results than does a census etc.

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STEPS IN SAMPLING DESIGN

- ❖ ***Type of universe:*** The first step in developing any sample design is to define the universe. Examples can be the population of a city, the number of workers in a factory, etc.
- ❖ ***Sampling Unit:*** A decision has to be taken concerning a sampling unit before selecting sample. Sampling unit may be a geographical one such as district, kebele, village, etc., or a social unit such as family, school, etc., or it may be an individual.
- ❖ ***Source list:*** It is also known as sampling frame from which sample is to be drawn.

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- ❖ *Size of sample*: This refers to the number of items to be selected from the universe to constitute a sample etc.

TYPES OF SAMPLE DESIGNS

- ❖ *Sample designs* are basically of two *types-Probability sampling and non-probability sampling*.
- ❖ *Probability sampling* is based on the concept of random/chance selection.
- ❖ Whereas *non-probability sampling* is *non-random selection*.

Probability sampling design

❖ *Probability sampling* is also known as random sampling or chance sampling. *Random sampling techniques* can be divided into simple random sampling and complex random sampling.

I. Systematic Sampling: In some instances, the most practical way of sampling is to select every *i^{th} item on a list*.

For example, if we want to take *20 samples* from a population of 100 members, our skip interval is 5 (i.e $100/20$). Our starting point must be selected randomly from the interval 1 to 5.

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II. Stratified Sampling:

- ❖ If a population from which a sample is to be drawn does not constitute a homogeneous group, stratified sampling is generally applied.
- ❖ Under stratified sampling, the population is divided into several *subpopulations (strata)* that are individually more homogeneous than the total population and then we select items from each stratum to constitute a sample.

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- ❖ To calculate sample size for each strata= $n(i/N)$
- ❖ For example, suppose we want to take a sample of size $n=30$ to be drawn from a population of size $N=800$ which is divided into three strata of size $N_1=400$, $N_2=240$, and $N_3=160$. The sample size for stratum with $N_1=400$ is $n_1=30(400/800)=15$. The sample size for stratum with $N_2=240$ is $n_2=30(240/800)=9$. The sample size for stratum with $N_3=160$ is $n_3=30(160/800)=6$.

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3. Cluster Sampling:

If the total area of interest happens to be a big one, a convenient way in which a sample can be taken is to divide the area into a number of smaller non-overlapping areas and then to randomly select a number of these smaller areas (clusters), with the ultimate sample consisting of all units in these small areas or clusters.

- ❖ *Cluster sampling* refers to a sampling method wherein the members of the population are selected at random, from naturally occurring groups called 'cluster'.

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- ❖ *Cluster sampling* is defined as a sampling technique in which the population is divided into already existing groupings (clusters), and then a sample of the cluster is selected randomly from the population.
- ❖ The most common variables used in the clustering population are the geographical area, buildings, school, etc. Heterogeneity of the cluster is an important feature of an ideal cluster sample design.
- ❖ *EXAMPLE:* In a survey of students from a city, we first select a sample of schools, then we select a sample of classrooms within the selected schools, and finally we select a sample of students within the selected classes. *READ MORE!!!!!!!!!!!!*

2. NON-PROBABILITY SAMPLING TECHNIQUES

In this type of sampling, items for the sample are selected deliberately by the researcher; his choice concerning the items remains supreme. Lets discuss the types :

- A. Judgment (purposive) sampling* - The researcher uses his judgment to select people that he feels are representative of the population to have a particular expertise or knowledge which makes them suitable.
- B. Convenience (accidental) sampling* - the most convenience population is chosen, which may be the researchers friends, work colleagues or students from a nearby college. This method is often used to save time and resources.

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- ❖ ***Quota sampling*** - The researcher selects a predetermined number of individuals from different group (*i.e. based on age, gender, etc*). This is perhaps the most popular non-probability sampling method used. Quota sampling is useful when a particular group or characteristics is relatively rare in the population.
- ❖ ***Referral Sampling***: this is a non-probability sampling technique which utilizes some form of referral, wherein respondents who are initially contacted are asked to supply the names and addresses of members of the target population.

TYPES AND SOURCES OF DATA

- ❖ Depending on the source, the type of data collected could be primary or secondary in nature.
- ❖ *Primary data* are those which are collected afresh and for the first time, and thus happen to be original in character. Its advantage is its relevance to the user, but it is also likely to be expensive in time and money terms to collect.
- ❖ *Secondary data* : is information extracted from an existing source, probably published or held on a computer database.

Methods of collecting primary data

- ❖ When existing data are not available for a particular population, it may be necessary to collect primary data.
- ❖ Primary data collection methods include, *but are not* limited to, *surveys, interviews, focus groups, and observations*. Many communities choose to utilize a mix of methods. This handout provides a broad overview of primary data collection methods.

1. Interview method

- ❖ *The interview method* of collecting data involves presentation of oral-verbal stimuli and reply in terms of oral-verbal responses.
- ❖ *Personal interviews*: this method requires a person (interviewer) asking questions in a face-to-face contact to the *interviewee*.
- ❖ *Telephone interviews*: this method of collecting information consists in contacting respondents on telephone itself.

2. Focus groups

are structured interviews with small groups of like individuals using standardized questions, follow-up questions, and exploration of other topics that arise to better understand participants.

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3. Observations

Another method of primary data collection involves observations. Observations can include community and/ or environmental scans, media scans, journals/logs, and photographs.

4. Questionnaire

❖ *Questionnaires* are data collecting instruments used to ask respondents to secure the desired information. It consists of a number of questions printed or typed in a definite order on a form or set of forms.

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- ❖ *The questionnaire* is mailed to respondents who are expected to *read and understand* the questions and write down the *reply in the space* meant for the purpose in the questionnaire itself.

The merits of this method are:

- It is free from the *bias* of the interviewer; answers are in respondents' own words
- Respondents have adequate time to give well thought out answers
- Respondents who are not easily approachable can also be reached conveniently.

Cont...

The main demerits of this system can be

- ✓ It can be used only when respondents *are educated* and *cooperating*
- ✓ The control over questionnaire may be lost once it is sent
- ✓ There is inbuilt inflexibility because of the difficulty of amending the approach once questionnaires have been dispatched/shipped.
- ✓ There is also possibility of *ambiguous replies* or omission of replies altogether to certain questions.

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The questions can be classified into two forms:

- ❖ *A closed type of question*, the respondent chooses between an alternative already stated. He does not get a chance to answer in a descriptive manner. For ex.: Do you use Brand X? Yes---- No----
- ❖ *In an open-ended question* the respondent is asked to give a reply to a question in his/her own words. *No* answers are suggested.
- ❖ These responses are *explanatory in nature* and give some insights from the respondents end.
- ❖ *Example: "What do you like most about this Product?"*

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Essentials of a good questionnaire:

- ✓ To be successful, *questionnaire* should be comparatively short and simple.
- ✓ *Questions* should proceed in *logical sequence* moving from easy to more difficult questions.
- ✓ *Personal questions* should be left to the end.
- ✓ Technical terms and vague expressions capable of different interpretations should be avoided.
- ✓ Don't use jargon/ abbreviations

Quiz (5%)

- 1. Define the research problem & research design.*
- 2. Write difference between sampling design & research design.*